

Refractive-Index Profile Dependence of an LPFG-Based Mode Scrambler for Broadband and Low-Loss MDL Compensation

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Abstract A dual-step-index (DSI) core reduces the insertion loss increase associated with mode-coupling efficiency for LPFG-type 2-LP mode scramblers. An LPFG-based mode-scrambler utilizing DSI-core fibre was numerically and experimentally demonstrated, achieving < 0.5 dB insertion loss and broadband mode-dependent-loss compensation over the C+L band. ©2026 The Author(s)

Introduction

Mode-division multiplexing (MDM) transmission using few-mode fibre (FMF) is considered a promising candidate to achieve significantly higher and denser optical transmission than conventional single-mode fibres [1]. In MDM systems, it is crucial to reduce the characterization differences among the propagation modes, such as mode-dependent loss (MDL) [2].

Mode-permutation and mode-scrambling schemes have been proposed to reduce the MDL in MDM systems [3,4]. While the mode-permutation scheme utilizing mode-(de)multiplexer (MUX/DEMUX) has achieved 6300-km-long MDM transmission [3], it tends to increase the repeater complexity and can lead to a higher insertion loss (IL). Long-period fibre grating (LPFG)-based mode scramblers have been reported to exhibit relatively low IL thanks to the good compatibility and connectivity with the transmission fibres [4]. In addition, lower differential mode delay (DMD) can enhance the operation wavelength bandwidth of LPFGs [5]. However, the dependence of LPFG-based MDL compensation devices on the refractive index profile of the host-fibre has not yet been clarified.

This paper examines the characteristics of mechanically induced LPFGs in two 2LP-FMFs, one with a graded-index (GI) core and one with a dual-step-index (DSI) core. Our findings reveal that a single LPFG-based mode scrambler with DSI fibre can achieve the broadband MDL compensation of about 1.5 dB over C + L bands.

Analytical Model

In this study, we examine the mechanically induced long-period grating shown in Fig. 1. When LP₀₁ and LP₁₁ modes satisfy the phase-matching condition $\Lambda = 2\pi / (\beta_{01} - \beta_{11})$, where Λ is a grating period and β_{01} and β_{11} are propagation constants of LP₀₁ and LP₁₁ modes, respectively, the mode-coupling amount of LP₀₁ to LP₁₁, C_{01} , can be expressed based on mode-coupling theory [6] as follows:

$$C_{01} = -10\log_{10} \left(1 - \sin^2 \left(\frac{\kappa_{01-11}L}{2} \right) \right), \quad (1)$$

where L is a grating length, and κ_{01-11} is the coupling coefficient between LP₀₁ and LP₁₁ modes. Here, we define the IL as the difference in total mode power output before and after the LPFG implementation, which arises from mode-coupling to cladding modes when the phase-matching condition between LP₀₁ and LP₁₁ is satisfied. We assumed that the propagation constant of any cladding mode is β_c and that the detuning parameter of the grating is $\delta = 2\pi / \Lambda - (\beta_{11} - \beta_c) = \beta_{01} - 2\beta_{11} + \beta_c$. The coupling amount α_c between LP₁₁ and a cladding mode can be calculated from the coupling coefficient κ_{11-cl} (the coupling coefficient between LP₁₁ and the cladding modes) and δ . Thus, the IL is obtained by summing α_c over all cladding modes as

$$IL = \sum \alpha_c = \sum 4.34 |\kappa_{11-cl}|^2 L^2 \text{sinc}^2 \left(\frac{\delta L}{2} \right). \quad (2)$$

Here, the coupling coefficient κ_{m-n} can be calculated from the overlap integral of the electric field

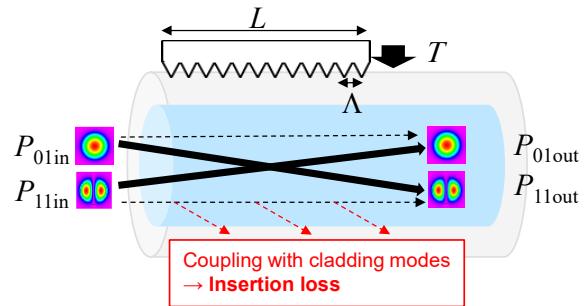


Fig.1: Schematic image of 2-LP mode scrambled and clad mode coupling induced IL occurring in mechanically induced LPFG.

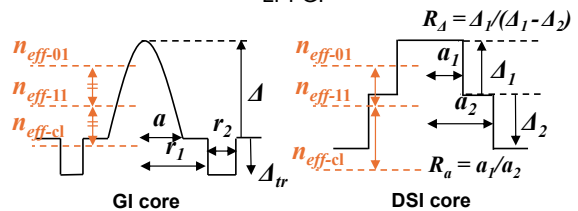


Fig. 2: Definition of refractive index profiles and image of the effective refractive indices for GI and DSI cores.

distributions and the refractive index modulation distribution Δn according to mode-coupling theory. We assumed anisotropic refractive index modulation in half of the core cross-section.

Figure 2 shows the structural definition of the refractive index profile and an image of the effective refractive indices for the GI and DSI cores [7,8]. For the GI core, the core radius, relative index difference, and alpha-parameter are $a = 10.3 \mu\text{m}$, $\Delta = 0.44\%$, and $\alpha = 2.4$, respectively. For the trench, r_1 , r_2 , and Δ_{tr} are $5 \mu\text{m}$, $10 \mu\text{m}$, and -0.3% , respectively. For the DSI core, $a_2 = 7 \mu\text{m}$, $\Delta_2 = -0.65\%$, $R_a = 0.35$, and $R_\Delta = 0.25$. $n_{\text{eff-01}}$, $n_{\text{eff-11}}$, and $n_{\text{eff-cl}}$ correspond to LP_{01} , LP_{11} , and the cladding modes, respectively.

In general, the n_{eff} difference between the LP_{01} and LP_{11} modes is similar to that between LP_{11} and the cladding modes in the GI core. Thus, the IL in the GI core with LPFG tends to be large [9]. In contrast, in the DSI core, the n_{eff} difference between LP_{11} and the cladding mode can be managed by designing R_a and R_Δ . Hence, it is expected that an LPFG-based mode scrambler inscribed in the DSI core can improve the IL compared to the GI core-based one.

Experiments and Discussion

We evaluated the optical characteristics and the bandwidth of the fabricated LPFG for the two fibre samples with the GI core and DSI core. Table 1 lists the measured DMD and the bandwidth over which the LP_{01} power after LPFG inscription is lower than -10 dB , $\lambda_{10\text{dB}}$, for GI and DSI, respectively. The grating period Λ was set to $723 \mu\text{m}$ for GI and $468 \mu\text{m}$ for DSI so that the coupling centre wavelength became 1550 nm . The transmission spectra were measured with an optical spectrum analyser utilizing a superluminescent diode (SLD) source, and LP_{01} was launched and received via planar lightwave circuit (PLC)-based mode MUX/DEMUXs at both ends of the 2LP-FMF. The bandwidths were evaluated from the spectral difference before and after LPFG inscription. The resulting 10-dB bandwidths, $\lambda_{10\text{dB}}$, were 231 nm and 78 nm for GI and DSI, respectively. Wide 10-dB bandwidths beyond C-band were observed in both samples, confirming that this is the effect of using low-DMD FMF, as expected from [5].

We then investigated the core profile dependence of IL and the mode coupling amount of the LPFG-based mode scrambler using two FMF samples. Since the LP_{11} output power after LPFG

Tab. 1: Measured optical characteristics.

	DMD [ns/km]	$\Delta\lambda_{10\text{dB}}$ [nm]
GI	0.02	231
DSI	-0.38	78

transmission, $P_{11\text{out}}$, is reduced by coupling to the cladding modes, it is difficult to estimate the coupling amount directly from $P_{11\text{out}}$. Therefore, based on the LP_{01} powers measured before and after LPFG insertion, $P_{01\text{in}}$ and $P_{01\text{out}}$, we defined $C_{01}' = P_{01\text{in}} - P_{01\text{out}}$ and regarded it as equivalent to C_{01} in (1). The IL was evaluated as the power attenuation caused by LPFG insertion, with all output modes collected using a bare connector.

Figure 3(a) shows the calculated results of C_{01} (dashed lines) and IL (solid lines) for the modulation depth Δn , where blue and red respectively indicate the GI and DSI cores. The wavelength λ and L were set to 1550 nm and 4 cm , respectively. IL was calculated as the sum of 20 modes near the cladding mode with the largest α_c , corresponding to the range in which the maximum coupling decreased by 20 dB or more in (2). We can see here that the increase in IL with increasing C_{01} is smaller in DSI than in GI. Owing to the relatively high design flexibility of its core profile, the DSI core achieve both LP_{01} - LP_{11} -specific DMD control and avoidance of phase-matching conditions to cladding modes, as shown in the right side of Fig. 2. For GI and DSI, the ILs were 3.5 dB and 0.4 dB , respectively, when $C_{01} = 10 \text{ dB}$ (i.e., the power coupling ratio from LP_{01} to LP_{11} was 90%). The increase in IL relative to the amount of coupling in GI-core fibre is calculated to be about 2.9 dB relative to DSI-core fibre.

Figure 3(b) shows the experimental results of C_{01}' (dashed lines) and IL (solid lines) for the

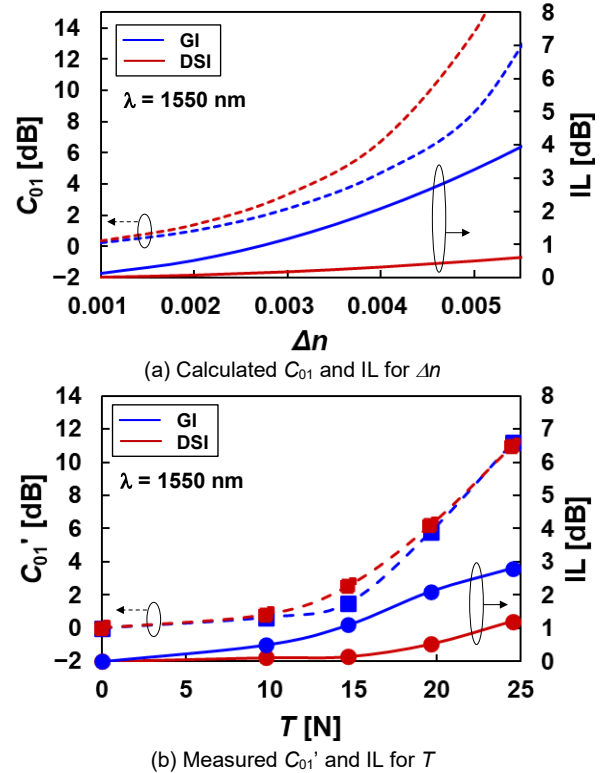


Fig. 3: Inter-mode coupling and insertion loss as a function of refractive index modulation.

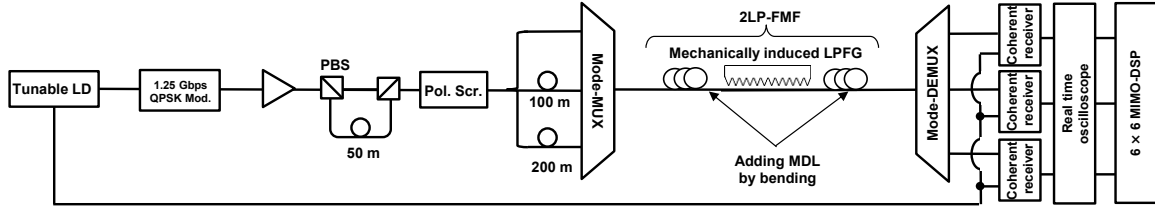


Fig. 4: Experimental setup for MDL measurement.

pushing amount T . Here, we induced the refractive index modulation by T . It is clear that an increase in IL associated with mode-coupling exhibits a trend comparable to the calculated results for both samples. These results demonstrate that LPFG in the DSI-core fibre exhibited lower IL than that in the GI-core fibre.

Next, we experimentally evaluated the MDL compensation performance in the transmission system with the two LPFGs, using the setup shown in Fig. 4. The output light of a tunable laser was modulated to 1.25 Gb/s using the QPSK, and polarization multiplexed/scrambled was utilized as the input signal. An input signal with an output level of -11 dBm was divided into three ports to be guided to each port of the mode-MUX and input to the sample with a GI core or DSI core, which are 2.5 m long. After separation by mode-DEMUX, MDL was evaluated using coherent 6×6 MIMO. Two bends were introduced to induce the MDL intentionally, resulting in a total MDL of 10 dB. FMF samples were placed at the centre of this configuration to evaluate the effect of LPFG.

Figure 5 shows the MDL changes, ΔMDL , relative to C_{01}' when the mechanically induced LPFG is implemented in the GI-core (blue dots) and DSI-core (red dots) fibres. Here, ΔMDL was defined as the difference between MDL values before and after LPFG implementation. The wavelength was set to 1550 nm. In the GI-core fibre, MDL worsened even when the coupling was increased due to the dominance of IL, as shown in Fig. 3. In contrast, IL had a smaller impact in the DSI-core fibre, where the MDL remained suppressed. The optimum condition appeared at $C_{01}' \approx 6$ dB, giving $\Delta\text{MDL} = -1.5$ dB. When C_{01}' exceeds 6 dB, IL grows and ΔMDL begins to rise. Here, based on the numerical model in [4], we performed calculations under conditions equivalent to those of the experimental environment and confirmed a similar reduction in MDL. These results demonstrate that C_{01} and IL exhibit a trade-off, and the LPFG coupling conditions should be selected appropriately according to the fibre structure and core design.

The blue and red dots in Fig. 6 show the wavelength dependence of ΔMDL after the LPFG was mechanically implemented in GI-core and DSI-core fibres, respectively. C_{01}' was fixed at 6 dB, and the number of averages was set to 100. Error bars indicate 3σ , where σ denotes the standard

deviation. Over the C+L band, the GI core showed a degradation of 2–3 dB due to IL increase, while DSI core showed an improvement of about 1.5 dB, confirming that there was sufficiently small wavelength dependence. These findings demonstrate the effectiveness of MDL compensation using a broadband, low-loss LPFG-based mode scrambler that utilizes a DSI-core fibre.

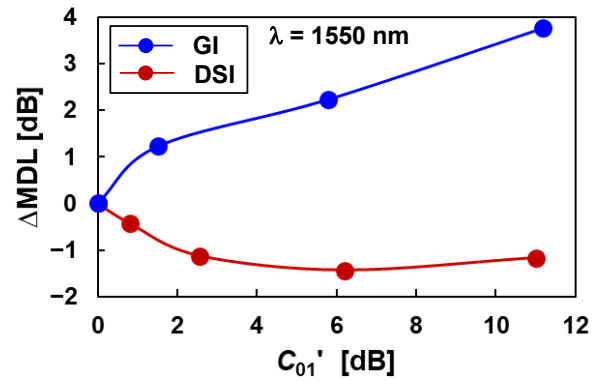


Fig. 5: Measured ΔMDL and IL for C_{01}' .

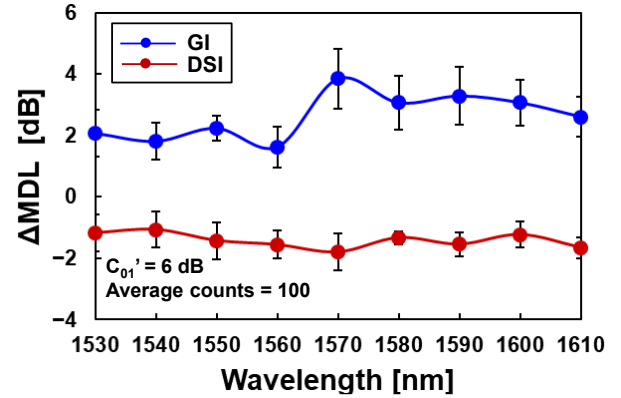


Fig. 6: Wavelength dependence of ΔMDL .

Conclusions

This study investigated the insertion loss (IL) characteristics of a 2LP-FMF LPFG-based mode scrambler with GI and DSI cores. We constructed an IL calculation model from the coupling between the LP_{11} mode and the cladding leakage modes, and experiments confirmed that the DSI core can suppress the increase in IL with increasing mode-coupling efficiency compared to the GI core. The single LPFG-based mode scrambler with a DSI core demonstrated an MDL compensation effect of approximately 1.5 dB over the C+L band.

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